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OPHTHALMIC MICROSURGICAL INSTRUMENT

Abstract

In some embodiments, a microsurgical instrument includes a trocar having a rigid, hollow shaft formed with a lumen extending from a proximal end to a distal end of the shaft. The distal end of the shaft may be shaped for tissue penetration. The instrument may further include a composite microcannula slidably engaged with the trocar in the lumen. The microcannula includes a light guide and a flexible hollow tube having an outer diameter less than an inner diameter of the lumen in the trocar. Other embodiments include placing the microcannula in the lumen of the trocar, illuminating the end of the trocar by illuminating the end of the microcannula, advancing the trocar from a selected entry point on an eye into a selected structure in the eye, and extending the illuminated end of the microcannula from the trocar into the selected structure.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of patent application Ser. No. 18/109,154, filed Feb. 13, 2023, which is a continuation of patent application Ser. No. 16/145,119, filed Feb. 9, 2018, which is a continuation-in-part of U.S. Pat. No. 10,987,247, issued Apr. 27, 2021, incorporated herein by reference in its entirety, which claims the benefit of U.S. Provisional Patent Application No. 62/574,136, filed Oct. 18, 2017, incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] Embodiments are related to surgical apparatus for treatment of eye diseases such as glaucoma.

BACKGROUND

[0003] Aqueous humor is a transparent, watery fluid produced within an eye, filling the anterior and posterior chambers of the eye, transporting agents needed by eye tissues and helping to maintain the eye's rounded shape by fluid pressure. Aqueous humor flows out of the eye through a fluid drainage network including the trabecular meshwork and Schlemm's canal, a porous, circumferential fluid passage coupled to collector channels and veins. Blockage or collapse of parts of the eye's drainage network can lead to an increase in intraocular pressure, a condition that may be associated with degraded vision and eye diseases such as glaucoma.

[0004] Surgical treatment may be used to reduce intraocular pressure by improving the flow of aqueous humor. Some surgical treatments involve making relatively large incisions through the sclera, the tough white outer covering of the eye, forming a flap of tissue folded back to expose the trabecular meshwork or other parts of the flow path for aqueous humor. Exposed parts of the drainage network may then be modified by removing tissue or forming new drainage channels. An incision through the sclera may lead to a loss of fluid pressure inside the eye and a collapse of one or more of the chambers. It may be necessary to support the eye's natural shape by injecting a viscoelastic fluid into one of the chambers. A viscoelastic fluid has a viscosity that changes from dynamic to static flow conditions, flowing with relatively low viscosity when subjected to shear stress and assuming a gel-like, high-viscosity state under static conditions.

[0005] Surgical treatments involving incisions into the eye may increase the risk of post-surgical complications such as infection and the formation of scar tissue. Other treatment procedures less disruptive to the eye have been developed. For example, the flow of aqueous humor may be improved by passing a microcannula through parts of the eye's drainage network to remove obstructions or re-open collapsed fluid passages. Additionally, it can be advantageous to deliver drugs or drug-eluting devices or materials into tissue structures. Notably, the delivery of drugs and drug-eluting devices to Schlemm's Canal can be beneficial due to Schlemm's canal being outside

the immune privileged area that is found inside the main body of the human eye. The microcannula may include a flexible hollow tube having an outer diameter small enough to permit the microcannula to be introduced into Schlemm's canal or some other part of the eye's drainage network. The microcannula may be sufficiently flexible to follow the curve of Schlemm's canal or another part of the drainage network while the microcannula is pushed from outside the eye, for example through a surgically-formed flap as previously described or through a puncture of the sclera. The microcannula may be used to mechanically expand a selected part of the drainage channel in the eye, or may be used to inject materials, objects, fluids, drugs, or viscoelastic to apply fluid pressure for improving flow through part of the eye's drainage system. Or, a microsurgical cutting, penetrating, or grasping instrument may be passed through the microcannula to guide the instrument to a part of the eye to be surgically modified.

[0006] Some microsurgical instruments have a microcannula slidably engaged with a hollow, flexible outer sheath. The flexible outer sheath may be used to position an entry point for the microcannula into the interior of an eye, with the microcannula passing through a lumen in the flexible outer sheath and the outer sheath held stationary with respect to the eye. An end of the microcannula may be extended from an end of the sheath to enter a selected part of an eye. However, the flexibility of the outer sheath may make it difficult for the sheath to penetrate the sclera or other tissue to enable the microcannula to enter drainage structures or other treatment areas inside the eye. It may be necessary to make an incision or puncture with a separate instrument to permit the flexible sheath to be positioned accurately for placement and guidance of the microcannula. Or, the microcannula may be provided with a tip shaped for tissue penetration, possibly limiting the use of the microcannula for delivering a payload into the interior of an eye.

SUMMARY

[0007] An example apparatus embodiment includes a trocar and a composite microcannula. An example of a trocar includes a rigid shaft having a proximal end and a distal end. The rigid shaft may be formed with a lumen extending from the proximal end to the distal end. The distal end of the rigid shaft on the trocar may be shaped for tissue penetration.

[0008] A composite microcannula may be positioned in the lumen of the trocar. An example of the composite microcannula may include a flexible hollow tube having an outer diameter less than an inner diameter of the lumen in the trocar. The example of a composite microcannula may further include a light guide.

[0009] Another example apparatus embodiment includes a trocar for ophthalmic surgery. The example of a trocar includes a rigid hollow shaft having a distal end shaped for tissue penetration. The hollow shaft is formed with a lumen extending from a distal end of the hollow shaft to a proximal end of the hollow shaft. The example of a trocar further includes a transition structure attached to the proximal end of the hollow shaft. The transition structure may be formed with an aperture for admitting a composite microcannula into the lumen. The example of a trocar may further include a light source disposed to illuminate the distal end of the rigid hollow shaft. A distal edge of the lumen in the hollow shaft may be smoothed to reduce abrasion and/or cutting of a solid object passing through the lumen and out of the trocar. The example of a trocar may further include a finger grip extending outward from the transition structure.

[0010] An example method embodiment includes placing a distal end of a composite microcannula within a lumen of a trocar; illuminating the distal end of the composite microcannula, thereby illuminating the distal end of the trocar; selecting a trocar entry point on an eye and positioning the trocar at the selected entry point; advancing the trocar at the selected entry point until the illuminated distal end of the trocar is observed to enter a selected structure in the eye; and extending the composite microcannula from the distal end of the trocar toward a target region in the eye, thereby transitioning from illuminating the distal end of the trocar to illuminating tissue outside the trocar.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a pictorial view of an example embodiment of a microsurgical instrument including a trocar with a rigid hollow shaft and a flexible composite microcannula.

[0012] FIG. 2 is a top view of an example of a trocar in accord with an embodiment.

[0013] FIG. 3 is a side view of the example of a trocar of FIG. 2.

[0014] FIG. 4 is a cross-sectional view A-A of the example of a trocar of FIGS. 1-3. A section line labelled A-A in FIG. 3 shows the position and viewing direction for the cross-sectional view in FIG. 4.

[0015] FIG. 5 is an enlarged partial cross-sectional view B of the rigid hollow shaft of FIGS. 1-4, showing an example of a distal end shaped for tissue penetration and further showing an example of a smoothed distal edge of the lumen in the trocar. The position of View B in FIG. 5 is marked with a broken line in FIG. 4.

[0016] FIG. 6 is a partial top view of an example of a composite microcannula in accord with the embodiments of the microsurgical instrument.

[0017] FIG. 7 is a cross-sectional view C-C of the example composite microcannula of FIG. 6. The position and viewing direction for cross-sectional view C-C in FIG. 7 is marked with a longitudinal section line C-C in FIG. 6.

[0018] FIG. 8 is a cross-sectional view D-D of the example composite microcannula of FIG. 6. The position and viewing direction for cross-sectional view D-D in FIG. 8 is marked with a transverse section line D-D in FIG. 6.

[0019] FIG. 9 is an alternative cross-sectional view D-D of another example of a composite microcannula in accord with embodiments of the microsurgical instrument. The position and viewing direction for alternative cross-sectional view D-D in FIG. 9 is marked with the transverse section line D-D in FIG. 6.

[0020] FIG. 10 is an alternative cross-sectional view D-D of another example of a composite microcannula in accord with embodiments of the microsurgical instrument. The position and viewing direction for alternative cross-sectional view D-D in FIG. 10 is marked with the transverse section line D-D in FIG. 6.

[0021] FIG. 11 is an alternative cross-sectional view D-D of another example of a composite microcannula in accord with embodiments of the microsurgical instrument. The position and viewing direction for alternative cross-sectional view D-D in FIG. 11 is marked with the transverse section line D-D in FIG. 6.

[0022] FIG. 12 is an alternative cross-sectional view A-A of another example of a trocar, further illustrating an example of a composite microcannula offset in a proximal direction from the distal end of the trocar, and further illustrating an example of an illuminated distal end of the trocar.

[0023] FIG. 13 continues the example of FIG. 12, showing an example of the illuminated distal end of the composite microcannula extending outward from the distal end of the lumen in the trocar.

[0024] FIG. 14 is an alternative cross-sectional view D-D showing another example of a composite microcannula with two light guides in the interior longitudinal void of the composite microcannula's flexible hollow tube.

[0025] FIG. 15 is an alternative cross-sectional view D-D showing another example of a composite microcannula with two light guides, one light guide in the interior of the composite microcannula's flexible hollow tube, and another light guide contacting an external surface of the flexible hollow tube.

[0026] FIG. 16 is an alternative cross-sectional view D-D showing another example of a composite microcannula with two light guides, one light guide in the interior of the composite microcannula's flexible hollow tube, and another light guide between the hollow tube and an outer sleeve.

[0027] FIG. **17** is an alternative cross-sectional view D-D showing another example of a composite microcannula with two light guides, one light guide in the interior of the composite microcannula's flexible hollow tube, and a second light guide held against the flexible hollow tube by an outer coating applied over the second light guide and the flexible hollow tube.

[0028] FIG. **18** is a block diagram of an alternative apparatus embodiment including a positioner for displacing the composite microcannula relative to the trocar, and optionally including a fluid injector for introducing fluid into the composite microcannula.

[0029] FIG. **19** shows a top view of an example of a microsurgical instrument embodiment with a positioner and a trocar.

[0030] FIG. **20** is a cross-sectional view E-E of the example of a positioner from FIG. **19**. A location and viewing direction for cross-sectional view E-E in FIG. **20** is marked with a longitudinal section line labeled E-E in FIG. **19**.

[0031] FIG. **21** shows a pictorial view of an example microsurgical instrument embodiment including a positioner with the distal tip of the trocar passing through the sclera of an eye into Schlemm's canal, illustrating an example of light emitted from the illuminated distal end of the trocar to accurately indicate the position of the trocar.

[0032] FIG. **22** shows a pictorial view with the example microsurgical instrument and trocar in the same position relative to Schlemm's canal as in FIG. **21**, and further illustrating an example of the illuminated distal tip of the composite microcannula extended outward from the trocar along the circumferential path of Schlemm's canal.

[0033] FIG. **23** is another alternative cross-sectional view A-A of the example of a trocar of FIG. **1**, showing an example of a composite microcannula with its light guide passing through the lumen of the trocar, and further showing positioned in the lumen a second, optional light guide separate from the light guide in the composite microcannula.

[0034] FIG. **24** shows a partial pictorial view of the example of a trocar with a composite microcannula and a second light guide as in the example of FIG. **23**, illustrating an example of the illuminated distal ends of the trocar and composite microcannula separated from one another along the circumferential path of Schlemm's canal.

[0035] FIG. **25** shows an alternative cross-sectional view D-D of an example of a composite microcannula embodiment carrying an example of an optional fluid payload and an example of an optional solid payload inside the flexible hollow tube.

[0036] FIG. **26** shows a pictorial view of an example of a trocar entry point marking instrument.

[0037] FIG. **27** shows a view toward examples of marking surfaces on marking pads on the trocar entry point marking instrument of FIG. **26**.

[0038] FIG. **28** shows a side view of the example trocar entry point marking instrument of FIGS. **26-27**.

[0039] FIG. **29** shows a view toward an example of the sclera and iris of a human eye, illustrating an example of a pattern of tangent lines formed by the trocar entry point marking instrument of FIGS. **26-28**.

[0040] FIG. **30** illustrates an example of some steps included in a method for inserting a trocar through the sclera of an eye and advancing a composite microcannula from the trocar into a structure such as Schlemm's canal.

[0041] FIG. **31** is a partial top view of another example of a composite microcannula in accord with the embodiments of the microsurgical instrument.

[0042] FIG. **32** is a cross-sectional view F-F of the example composite microcannula of FIG. **31**. The position and viewing direction for cross-sectional view F-F in FIG. **32** is marked with a longitudinal section line F-F in FIG. **31**.

[0043] FIG. **33** is a cross-sectional view G-G of the example composite microcannula of FIGS. **30-31**. The position and viewing direction for cross-sectional view G-G in FIG. **33** is marked with a transverse section line G-G in FIG. **31**.

[0044] FIG. 34 is a view toward a top side of another an example of a microsurgical instrument embodiment with a positioner and a trocar.

[0045] FIG. 35 is a side view of two of the microsurgical instrument examples of FIG. 34, with the upper device showing the composite microcannula in an example of a retracted position and the lower device showing the composite microcannula in an example of an extended position.

[0046] FIG. 36 continues the example of FIGS. 34-35, showing the examples of a microsurgical instrument in cross-sectional view H-H for the retracted position of the composite microcannula and cross-sectional view K-K for the extended position of the composite microcannula. The positions of cross sections H-H and K-K are marked by section lines H-H and K-K in FIG. 35.

DESCRIPTION

[0047] Example embodiments in accord with the invention are described herein. A trocar having a rigid hollow shaft with a distal end shaped for tissue penetration is configured to pierce biological tissue, for example the sclera of an eye, forming a very small entry point into the tissue for a composite microcannula passing through a lumen of the trocar. The composite microcannula, which may also be referred to herein as a composite microcatheter, includes a light guide for illuminating the distal end of the microcannula. A light source may be coupled to the composite microcannula and/or trocar to illuminate the distal end of the rigid hollow shaft by directing light through the light guide into the lumen of the trocar, thereby enabling an accurate determination of the position of the distal end of the trocar, a visual indication of the entry of the trocar a structure in the eye, and a visual indication of the position of the distal end of the composite microcannula as it advances toward a target region for treatment. The illuminated distal end of the composite microcannula may be used to determine when the microcannula has deviated from a preferred path, for example leaving a preferred path through Schlemm's canal and entering another channel or chamber such as a collector channel or another part of the eye's drainage system.

[0048] Embodiments are effective for providing a visual indication of the location of the distal end of the trocar by observing light emitted from the trocar through tissue which may include the sclera, trabecular meshwork or other tissue, including tissue not associated with an eye. Light passing through the sclera or other tissue from the trocar further indicates a direction of travel of the trocar. The position and direction of travel of the composite microcannula may also be accurately determined by visual observation of the light emitted from the tip of the microcannula. An embodiment may be accurately guided into tissues and/or tissue spaces such as, but not limited to, the trabecular meshwork, Schlemm's canal, and collector channels. Conversely, an embodiment may be accurately guided to specifically avoid entering a selected tissue or tissue space. The tissues in the eye and the position and direction of travel of an embodiment may be observed directly, with a camera, with a gonioscope, with other optical aids, or any combination of these devices and methods.

[0049] In some embodiments, a second light guide enables independent, and optionally simultaneous, illumination of the distal ends of the trocar and the composite microcannula. In other embodiments, a payload comprising a fluid and/or a solid object may be delivered through the composite microcannula to a target region in an eye. Examples of a fluid payload include, but are not limited to, drugs including gene therapy, stem cell, and other fluid-based drugs, a viscoelastic fluid, water, and a saline solution. Examples of a solid payload include, but are not limited to, devices, particles, nano particles, small devices including drug-eluting examples of solid payloads, a microsurgery instrument such as forceps, an instrument for penetrating and/or cutting tissue, a stent, a light guide, and a wire. As used herein, a light guide refers to an optical element capable of transmitting electromagnetic energy received at an input surface to an output surface through an intervening optical medium. Examples of a light guide include, but are not limited to, one or more mirrors arranged to direct a light beam from a source to a destination, a flexible optical fiber, a bundle of optical fibers, and a rigid light pipe.

[0050] Some embodiments include a positioner for displacing the composite microcannula relative

to the distal end of the trocar. A positioner may optionally include a microcannula displacement mechanism configured to extend and optionally retract the composite microcannula. The positioner may further optionally include a fluid injector configured to move fluid from a fluid reservoir into the composite microcannula and possibly into a selected target region in an eye. Some embodiments of a positioner include a light source disposed to emit light into the light guide of the composite microcannula and optionally into a second light guide coupled to the trocar, when a second light guide is provided. A positioner may enable accurate advancement and/or retraction of the composite microcannula without disturbing the microcannula's entry point into the eye, possibly reducing an amount of time needed to complete a treatment procedure and reducing a risk of damage to eye tissue.

[0051] Embodiments of an ophthalmic microsurgical instrument may be configured to smoothly and continuously transition from illuminating the distal end of the trocar to illuminating tissue outside the trocar, thereby enabling very precise determination of the position of the trocar and composite microcannula relative to structures in the eye. The very small puncture made by the trocar in the sclera or other parts of the eye contrasts with the relatively large incisions required by previous surgical techniques that raise a flap of tissue from the sclera to access structures in the interior of the eye. The small puncture reduces patient discomfort and risk of post-operative complications such as scarring and infection. Preparing, monitoring and closing the surgical area are faster and less complicated than methods using incisions through the sclera, possibly enabling embodiments to be employed with less stringent levels of sterility and patient monitoring than may be practiced in an operating room for surgical procedures, and possibly permitting more rapid patient recovery and healing from surgical procedures.

[0052] An example embodiment of a microsurgical instrument appears in FIG. 1. The example embodiment **100** includes a trocar **200** configured to receive a composite microcannula **300**. The composite microcannula **300** may be slidably engaged with the trocar **200**, passing through a trocar connector **222** at the proximal end **204** of the trocar, a transition structure **214**, and a lumen **208** formed in a rigid hollow shaft **206** extending outward from the transition structure **214** to the distal end **202** of the trocar **200**. The rigid hollow shaft **206** is preferably formed with a distal end **210** shaped for tissue penetration. One or more of an optional finger grip **216** may be attached to, or alternately formed as an integral part of, the trocar connector **222** and/or the transition structure **214**.

[0053] In the example of FIG. 1, the composite microcannula **300** is shown with several bends and curves to illustrate the flexibility of the hollow tube **302** forming much of the length of the composite microcannula. The flexibility of the composite microcannula enables the microcannula to follow the curved walls of a structure in the eye, for example Schlemm's canal, without puncturing or damaging the walls of the structure. An embodiment of the composite microcannula **300** may be formed with segments of the microcannula having flexural rigidity in a range from 3.0×10^{-11} kN-m to 2.9×10^{-10} kN-m. Parts of the composite microcannula near the distal and/or proximal ends may optionally be more rigid than other parts of the composite microcannula. The rigid hollow shaft **206** of the trocar **200** is substantially stiffer than the composite microcannula and is preferably formed with a flexural rigidity of at least 1.5×10^{-8} kN-m, stiff enough to readily penetrate the sclera and other tissue in an eye. A trocar with flexural rigidity greater than the preferred minimum value may eliminate the need for a separate surgical instrument to form a puncture through an outer surface of an eye.

[0054] The composite microcannula **300** may include an optional microcannula connector **314** at the proximal end **310** of the flexible hollow tube **302**. The microcannula connector may include connections for introducing a payload into the composite microcannula and for coupling light from a light source into the composite microcannula. A liquid, solid, or gaseous payload introduced into the proximal end **310** may be transported through the hollow tube **302** to the distal end **308** of the composite microcannula for delivery to a target region in an eye. Light incident on the proximal

end **310** may travel to the distal end **308** to create an illuminated distal end **326** of the composite microcannula. An optional light diffuser **311** may be provided at the distal end **308** to disperse light in many directions, thereby indicating the precise location of the distal end of the composite microcannula as it moves through channels and chambers in an eye. Light may travel from the proximal end to the distal end of the composite microcannula by internal reflection from the walls of the flexible hollow tube **302**, through a liquid introduced into the hollow tube **302**, or through one or more light guides included with some embodiments of a composite microcannula.

[0055] An example embodiment of a trocar **200** is shown in a top view in FIG. 2 and a side view in FIG. 3. In the example of FIG. 3, the optional finger grips **216** visible in FIG. 2 have been omitted. The trocar connector **222** at the proximal end **204** of the trocar **200** may be a luer connector, for example a slip-fit or twist-lock luer connector. Other connectors capable of forming a leak-resistant seal may alternatively be used. The rigid hollow shaft **206** of the trocar **200** attaches to the transition structure **214**. A lumen **208** in the rigid hollow shaft **206** is in fluid communication with void spaces in the transition structure and trocar connector, thereby enabling fluid to be introduced into the lumen **208**. The lumen **208** of the trocar **200** extends through the distal end shaped for tissue penetration **210**.

[0056] Cross-sectional view A-A in FIG. 4 and partial enlarged view B in FIG. 5 illustrate some internal details of an example of a trocar **200** in accord with an embodiment. The rigid hollow shaft **206** is held firmly by the transition structure **214** attached to the trocar connector **222**. The lumen **208** through the rigid hollow shaft **206** is in fluid communication with void spaces **224** in the trocar connector **222**. The void space **224** may be formed with a conical microcannula guide surface **221** near the proximal end of the hollow shaft **206**. The conical surface **221** may deflect a composite microcannula toward the lumen **208** through an aperture **223** formed in the transition structure **214** near the distal end of the void space **224**.

[0057] The distal edge **212** of the lumen **208** is preferably smoothed, for example by rounding the edge **212** all the way around the distal end of the lumen. The smoothed distal edge **212** reduces abrasion or cutting of material from the composite microcannula when the microcannula slides across the distal edge of the trocar lumen. If left unsmoothed, the distal edge of the trocar lumen may be sharp enough to remove material from the composite microcannula. Reducing an amount of material cut or abraded from the composite microcannula reduces undesirable deposition of such material in an eye.

[0058] View B in FIG. 5 further illustrates an example of an outer diameter **219** of the rigid hollow shaft **206** and an inner diameter **218** of the lumen **208** through the rigid hollow shaft **206**. The outer diameter **219** may be in a range from about 200 microns to about 700 microns. For example, some trocar embodiments have a rigid hollow shaft with an outer diameter **219** of 450 microns. Other trocar embodiments have a rigid hollow shaft with an outer diameter **219** of 250 microns. The inner diameter **218** is preferably larger than a largest transverse dimension **306** of a composite microcannula configured to pass slidably through the lumen, for example an outer diameter **306** of the flexible hollow tube **302**, a largest transverse dimension **306** across the tube **302** and an outer coating **321** applied to the tube, a largest transverse **306** dimension across the hollow tube **302** and an external light guide **304** in contact with the tube, or an outer diameter **306** of a sleeve **320** surrounding the tube **302**.

[0059] An example embodiment of a composite microcannula **300** is shown in a top view in FIG. 6, a longitudinal cross-sectional view C-C in FIG. 7, and alternative transverse cross-sectional views D-D in FIGS. 8-11. As suggested in FIGS. 6, 7, and 8, light guide **304** may be positioned in a longitudinal void **303** extending from the proximal end **310** to the distal end **308** of the flexible hollow tube **302**. The longitudinal void **303** inside the composite microcannula **300** may also be referred to as the lumen **303** of the composite microcannula. The void **303** may serve as a fluid path **324** for a fluid introduced into the flexible hollow tube **302**. A payload introduced onto the composite microcannula may follow the fluid path **324** as the payload moves from the proximal

end **310** to the distal end **308**.

[0060] The light guide **304** may be formed separately from the flexible hollow tube as suggested in the previous examples. Alternatively, a light guide may be formed as an internal layer of the flexible hollow tube **302** as shown in the examples of FIGS. **31**, **32**, and **33**. The light guide **304** may be disposed as a concentric layer of material adjacent the void **303** in the composite microcatheter **300**. An index of refraction of the material of the light guide **304** preferably differs sufficiently from an index of refraction of the material of the flexible hollow tube **302** to enable efficient coupling of light from a light source to the distal end **308** by internal reflection through the light guide. The layer of material forming the light guide **304** may be formed by molding, chemical deposition, or by mechanically inserting a hollow tube into the flexible hollow tube **302**. Although the figures show an example of a light guide made from a single layer of material, the light guide may alternatively be made of several layers of material, each with a selected value of refractive index, or may alternatively be made with a refractive index that varies with distance from an edge of the light guide.

[0061] FIGS. **31**, **32**, and **33** further illustrate an example of a light diffuser **311** having a reduced outside diameter **312** compared to the outside diameter of the light diffuser **311** in the examples of FIGS. **6** and **7**. The light diffuser **311** may alternatively be formed as a rounded end of the flexible hollow tube **302**, where the diffuser has a radius equal to half the diameter **313** of the flexible hollow tube.

[0062] A second light guide and/or other liquid or solid payloads may be passed through the void **303** surrounded by the light guide **304**. In some embodiments, the light guide **304** entirely surrounds the void space **303** in the flexible hollow tube **302**. Alternatively, the light guide may be not entirely surround the void space, for example being formed as a hollow tube split longitudinally in half, in quarter, or some other fraction of a complete hollow tube.

[0063] An outer diameter **306** of the composite microcannula **300** may be the largest diameter found on the composite microcannula, for example the diameter **312** in FIG. **6**. The outer diameter **306**, for example the larger of the outside diameter **312** of an optional light diffuser **311** at the distal end **308** and the outside diameter **313** of the hollow tube **302**, is preferably less than the internal diameter **218** of the lumen **208** in the trocar **200**.

[0064] The distal end **308** of the composite microcannula corresponds to the illuminated end **326** when light incident on the proximal end **310** of the light guide **304** is emitted from the distal end **308**. As suggested in FIGS. **6** and **7**, the distal end **308** of the composite microcannula may be rounded to reduce tissue trauma and to disperse light from the light diffuser **311** in many directions to enhance the visibility of the illuminated distal end **326**. The outside diameter **312** of the light diffuser may be larger than the outside diameter **313** of the hollow tube **302**. Alternately, the diameters (**312**, **313**) may be approximately equal to one another.

[0065] For some embodiments, a length of a flexible segment **322** of the composite microcannula **300** may be a few millimeters longer than a circumferential length of Schlemm's canal. The circumferential length of Schlemm's canal in a human eye is about 36 millimeters. For some embodiments of a composite microcannula **300**, the length of the flexible segment **322** may be greater than 40 millimeters (1.6 inches). A length of the flexible segment **322** may optionally be substantially longer than the circumferential length of Schlemm's canal, for example to permit the composite microcannula to connect to a light source or fluid injection apparatus or to provide a convenient length outside the proximal end of the trocar for gripping the composite microcannula with forceps or fingers.

[0066] Alternatively, the length of the flexible segment **322** of the composite microcannula **300** may be about 20 millimeters, allowing catheterization of Schlemm's canal in two passes. A first pass may proceed through about half of the length of Schlemm's canal in a clockwise direction. A second pass may proceed through the other half of Schlemm's canal in a counterclockwise direction.

[0067] In the example of a composite microcannula **300** in FIG. **8**, the light guide **304** is positioned within the longitudinal void **303**. The light guide has a smaller diameter than the inner diameter **307** of the flexible hollow tube **302**. The outer diameter **306** of the composite microcannula **300** may be the outer diameter of the hollow tube **302**.

[0068] In the example of a composite microcannula **300** in FIG. **9**, the light guide **304** is positioned on the outside surface of the flexible tube **302**, leaving the space inside the flexible tube **302** available for transporting a payload through the composite microcannula. In the example of FIG. **9**, the outer diameter **306** of the composite microcannula includes the dimensions of the flexible tube **302** and the light guide **304**.

[0069] FIGS. **10** and **11** show more examples of a composite microcannula **300** in accord with a microsurgical instrument embodiment **100**. The example composite microcannula **300** in FIG. **10** positions the light guide **304** outside the void **303** in the flexible hollow tube **302**. An outer sleeve **320** surrounds the light guide **304** and the flexible tube **302**. The outer diameter **306** of the composite microcannula may correspond to the outer diameter of the sleeve **320** in FIG. **10**. In FIG. **11**, an example of a coating **321** has been applied over the light guide **304** and flexible hollow tube **302**. The outer diameter **306** along the flexible segment of the composite microcannula includes the dimensions of the hollow tube **302**, light guide **304**, and outer coating **321**. The example embodiments in FIGS. **10** and **11** both offer the full internal diameter **307** of the hollow tube **302** for carrying a payload through the longitudinal void **303**.

[0070] The composite microcannula may be positioned inside the lumen of the trocar to illuminate the distal end of the trocar while the trocar is being inserted into an eye. FIGS. **12** and **13** show an example of an alternative configuration for a transition structure **214** and examples of positions of the composite microcannula. In FIG. **12**, the composite microcannula is positioned for lighting the distal end **202** of the trocar's rigid hollow shaft **206**. The distal end **308** of the composite microcannula, which in the examples of FIGS. **12-13** is also the illuminated end **326** of the microcannula, may be installed in the lumen of the trocar a preferred offset distance **626** from the distal end **202** of the trocar. Internal reflection in the trocar lumen causes light **614** emitted from the distal end of the composite microcannula to emerge from the distal end of the trocar as light **612** dispersed across a wide range of angles, making the trocar tip visible from many different viewing directions. The illuminated distal end **220** of the trocar may be used to identify the precise location of the tip of the trocar inside eye tissue. For example, a readily perceived change in the brightness, as may be observed through the trabecular meshwork gonioscopically, of the emitted light **612** indicates when the distal end of the trocar has passed through the sclera and entered Schlemm's canal.

[0071] Advancing the illuminated distal end **326** of the composite microcannula out of the lumen in the trocar causes a smooth transition from illuminating the tip of the trocar to illuminating tissue outside the trocar. In the example of FIG. **13**, light **614** emitted from the illuminated distal end **326** of the composite microcannula may be dispersed across a wide range of angles, accurately indicating the position of the tip of the composite microcannula inside an eye.

[0072] Some alternative embodiments of a composite microcannula include two light guides as shown in the examples of FIGS. **14-17**. In the example of a composite microcannula **300** in FIG. **14**, a first light guide **304** and a second light guide **412** are disposed within the lumen **303** of the hollow tube **302**. In the example of FIG. **15**, the first light guide **304** may be outside the lumen **303** of the hollow tube **302**, and the second light **412** may be inside. In the example of FIG. **16**, both light guides may be positioned as in FIG. **15**, with an outer sleeve **320** surrounding the flexible hollow tube **302** and the first light guide **304**. In the example of FIG. **17**, both light guides may again be positioned as in FIG. **15**, but an outer coating **321** is applied over the first light guide **304** and flexible hollow tube **302**.

[0073] Some embodiments of a microsurgical instrument include a positioner for displacing the composite microcannula relative to the trocar. A block diagram of an alternative embodiment of a

microsurgical instrument **100** with a positioner is shown in the example of FIG. **18**. The positioner **400** may include a handpiece **420** holding a microcannula displacement mechanism **425** configured to extend, and optionally retract, the composite microcannula **300** from the trocar **200**. The trocar **200** may be attached to a trocar receiver **422**, for example a receiver for the luer fitting on the trocar. The composite microcannula may pass through a hollow sleeve **433** disposed between the trocar receiver **422** and the microcannula displacement mechanism **425**. The hollow sleeve **433** may improve smooth extension and retraction of the composite microcannula by reducing buckling or kinking of the flexible part of the composite microcannula inside the positioner **400** when the microcannula displacement mechanism is operated. An actuator **424** mechanically linked to the microcannula displacement mechanism **425** enables manual control of the length of the composite microcannula extended from the distal end of the trocar.

[0074] An optional light source **328** may be provided inside the positioner **400**. Light output from the light source **328** may be coupled into the microcannula connector **314** attached to the composite microcannula **300**. The microcannula connector **314** may optionally be configured to receive light from an external light source **330**. In some embodiments, the light source **328** may be disposed to transmit light through the second light guide **412** into the lumen of the trocar **200**, possibly through the intervening trocar receiver **422**.

[0075] The microcannula connector **314** may optionally provide a fluid connection to a fluid injector **446** disposed to transfer fluid from a fluid reservoir **442** inside the handpiece **420** to the composite microcannula **300**. The microcannula connector **314** may alternatively be connected to an external fluid injector **448** configured to move fluid from an external fluid reservoir **444** into the composite microcannula **300**.

[0076] FIGS. **19** and **20** show some details of an example of a microsurgical instrument embodiment **100** including a positioner **400**. The trocar connector **222** at the proximal end of the trocar **200** connects to the trocar receiver **422** at the distal end **418** of the positioner **400**. The trocar receiver **422** may be formed as an integral part of the handpiece **420**, or may alternately be formed separately and strongly attached to the handpiece. An actuator **424** slidably engages with the handpiece **420** along an actuator aperture **428** to extend the composite microcannula from the trocar. The actuator may be attached to, or alternately formed as an integral part of, a guide block **426** configured to track along a guide ridge **430** inside the handpiece **420**.

[0077] A hollow sleeve **433** may be connected at its proximal end to the guide block **426** and at its distal end to the trocar **200**. A composite microcannula (not visible in FIGS. **19-20**) may be positioned inside the hollow sleeve **433**. The hollow sleeve may include a fixed segment **436** attached to, or alternately formed as an integral part of, the trocar **200**, and a movable segment **434** connected to the actuator block **426**, where the length **438** of the hollow sleeve **433** includes both the fixed and movable segments. The fixed and movable segments of the hollow sleeve **433** may alternately be implemented as a hollow, collapsible and extendable bellows sleeve with concertinaed sides. The length **438** of the hollow sleeve **433** may change with a displacement of the composite microcannula **300** relative to the trocar **200**.

[0078] An end cap **440** may close off the proximal end **416** of the positioner **400**. The end cap **440** may be formed with a cap aperture **482** to permit the composite microcannula to extend out from the proximal end **416**.

[0079] FIGS. **21** and **22** illustrate an example of a location of a trocar entry point on an eye and extension of the composite microcannula to follow the circumferential path of Schlemm's canal. FIGS. **21** and **22** both show a pictorial view of the positioner **400** in a same position and orientation relative to an eye **1000**. Other parts of the approximately spherical eye **1000** shown in simplified form in the figures include the sclera **1002**, the iris **1006**, the pupil **1016**, and the circumferential path of Schlemm's canal **1010** near the limbus of the eye. Schlemm's canal **1010** includes a porous, approximately circular drainage channel receiving aqueous humor flowing through the trabecular meshwork near the outer edge of the iris. Schlemm's canal is marked in the figures with a hidden

line to represent the canal's approximate position behind the exterior surface of the sclera **1002**. The cornea, extending outward toward the viewer above the iris **1006**, may be considered to be present in FIGS. **21** and **22**, but is transparent and is not marked. A center **1018** of the pupil **1016** also indicates the approximate center of the iris **1006**.

[0080] In the example of FIGS. **21** and **22**, the distal end **202** of the trocar **200** has been inserted into the sclera **1002** at a preferred trocar entry point **606**, puncturing the sclera with a small hole having a diameter about the same as the outer diameter of the rigid hollow shaft **206**. The trocar may be advanced in a direction **610** along a line **604** tangent to Schlemm's canal **1010**. In the example of FIG. **21**, the distal end **202** of the trocar **200** is positioned as having just entered the interior of Schlemm's canal **1010**. The position of the distal end of the trocar is made visible by light **612** radiating from the lumen of the trocar. In some embodiments **100**, the light **612** will have been emitted from the distal end of a light guide with the end of the light guide offset a selected distance **626** from the distal end of the trocar lumen **208** (ref. FIG. **12**). Light **612** radiating from the interior of Schlemm's canal through the sclera may have the visual appearance from outside the eye of a bright spot on the exterior of the eye, and this light radiating from the interior of Schlemm's Canal through the Trabecular Meshwork may have the visual appearance of a well-defined bright spot visible across the anterior chamber of the eye. The bright spot is a visual indication of the precise position of the distal end **202** of the trocar **200**. Light **612** emitted from the end of the trocar and light **614** emitted from the end of the composite microcannula are represented by short wavy lines in the figures. A gonioscope may be used to view the bright spots marking the positions of the trocar and composite microcannula.

[0081] An example of a preferred entry point **606** of the trocar **200** through the sclera is shown in FIG. **21** along a line **604** tangent to Schlemm's canal. The entry point **606** of the trocar **200** into the sclera **1002** may be offset a predetermined distance from the limbus on the tangent line **604**. The offset distance may be selected to cause the trocar to pass into the interior of Schlemm's canal when inserted at the preferred entry point **606** with the shaft **206** of the trocar parallel to the tangent line. A trocar entry point marking instrument may be used to mark the surface of an eye with the preferred entry point **606** and direction of advance for the trocar along a tangent line **604**, as will be explained in more detail with regard to FIGS. **26-29**.

[0082] The actuator **424** on the example of a positioner **400** is shown near the proximal end of the actuator's range of travel in FIG. **21**. As the actuator is moved from its position in FIG. **21** in a distal direction to the position shown in FIG. **22**, the microcannula displacement mechanism **425**, which may include the examples of the actuator block **426** and the hollow sleeve **433** from the positioner embodiment **400** of FIG. **20**, causes the composite microcannula to extend outward from the distal end of the trocar. The example of a displacement distance **630** in FIG. **22** corresponds to a length of the portion of the composite microcannula extending outward from the distal end of the trocar by movement of the actuator **424**.

[0083] In the example of FIG. **22**, the segment of the composite microcannula extending outward from the distal end **202** of the trocar **200** follows the circumferential path of Schlemm's canal **1010** in a counterclockwise direction **611** from the trocar entry point **606**. Light **614** emitted from the distal end of the composite microcannula **300** may be visible through the sclera as a small spot of light, accurately indicating the position of the distal end of the composite microcannula. The composite microcannula could alternately be made to follow Schlemm's canal in a counterclockwise direction by reorienting the positioner **400**. Should the composite microcannula deviate from a preferred path, for example leaving Schlemm's canal and entering a collector channel, the path change will quickly become apparent because of the illuminated distal tip of the microcannula.

[0084] The positioner **400** may remain stationary relative to the eye **1000** while the composite microcannula moves through Schlemm's canal or other parts of the eye. The positioner **400** may hold the composite microcannula stationary relative to the eye, for example while a payload is

being delivered through the composite microcannula to a target region in the eye. Although the examples of a trocar in FIGS. **19-22** do not include finger grips, the examples apply also to a trocar with a finger grip. A finger grip may be used to immobilize the positioner, for example by holding the finger grip to a patient's skin with adhesive tape or a temporary stitch.

[0085] Some embodiments of a trocar are configured to receive two light guides. One of the light guides may be included in a composite microcannula as previously described. The second light guide may be in another composite microcannula or may be provided independently of a composite microcannula. Examples of a trocar adapted for two light guides are shown in FIGS. **23** and **24**. In the example of FIG. **23**, the composite microcannula **300** is shown with its distal end **308** extending out from the distal end **202** of the trocar **200**, and with light **614** being emitted from the light guide in the composite microcannula. An optional second light guide **412** is also positioned in the lumen of the trocar **200**, with the distal end **413** of the second light guide **412** remaining inside the trocar, radiating light **612** from the distal end of the trocar. As suggested in FIG. **24**, the two light guides may optionally selectively illuminate the distal end **202** of the trocar **200** only, the distal end **326** of the composite microcannula only, or both, either sequentially or simultaneously. The emitted light (**612**, **614**) may optionally have a wavelength and/or intensity not visible to unaided human vision.

[0086] An optional camera **616** may be provided to capture images of the emitted light (**612**, **614**) passing through eye tissue from the trocar and/or microcannula. The positions of the composite microcannula and trocar may be visible in an image from the camera **616** presented on a computer monitor, smart phone display, and/or instrument display. In some microsurgical instrument embodiments **100**, the composite microcannula **300** and the second light guide **412** may both receive light from the same light source. The composite microcannula may be advanced until it reaches a target region **1014**, for example an area to be cleared of an obstruction or constriction, or an area that will receive a payload delivered through the composite microcannula.

[0087] FIG. **25** shows an example of a payload **620** carried in the void **303** inside the flexible hollow tube **302** of a composite microcannula **300**. The payload may follow the fluid path **324** through the composite microcannula. The composite microcannula may therefore be used to precisely position the payload in a target region of an eye. The payload **620** may be a solid object **624**, a fluid **622**, for example a fluid comprising a gas and/or a liquid, or both a solid object and a fluid. The fluid may optionally be used to transport a solid object **624**, or a long solid payload may be pushed in from the proximal end of the composite microcannula until the payload extends out the distal end of the composite microcannula. The example of a composite microcannula **300** in FIG. **25** has a light guide **304** inside the flexible hollow tube. Other embodiments of a composite microcannula **300** disclosed herein may also be used to deliver a payload **620**.

[0088] FIGS. **26**, **27**, and **28** show an example of a trocar entry point marking instrument **800**, also referred to as a marking jig **800**. The marking instrument may be used to form a pattern of tangent lines on the sclera of an eye. The pattern of lines marks at least one, and optionally more than one, preferred trocar entry point and preferred trocar insertion direction for guiding a trocar **200** through the sclera into the interior fluid passage in Schlemm's canal. The marking instrument **800** includes at least one pair of marking pads **810**, **812**. Each marking pad **810**, **812** is preferably positioned to form a line tangent to Schlemm's canal when an edge **808** of a sighting aperture through a hub **806** is concentric with the pupil of the eye. A dye applied to a contact surface **814** on each marking pad **810**, **812** may be transferred to the sclera **1002** as intersecting line segments when the contact surfaces touch the surface of the eye.

[0089] The two marking pads **810**, **812** in each pair are disposed at an angle to one another such that the intersection of the two line segments formed on the eye marks the location of the preferred trocar entry point **606**. The intersection point of the two line segments is preferably offset by a predetermined separation distance **826** from the limbus **1008**, measured in a radial direction from the center **1018** of the pupil. The limbus **1008** indicates the underlying position of Schlemm's canal **1010** with sufficient precision for the markings made by the two pads **810**, **812** to accurately

indicate the insertion position and direction of the distal end of the trocar for entry into Schlemm's canal. The predetermined separation distance may be determined from a selected length of each line segment to be marked on the eye and from the number of separate trocar entry points **606** to be marked on the eye.

[0090] The pairs of pads **810**, **812** may be connected to a central hub **806**. A handle **802** may be attached to the central hub **806**. The pads **810**, **812** may be connected directly to the hub, or may alternatively be connected to the hub by an intervening radial arm **804**. The example of a trocar entry point marking instrument **800** in FIG. **26** includes a hub **806** with seven radial arms **804**. A first marking pad **810** and a second marking pad **812** disposed at an angle to the first marking pad are connected to each radial arm **804**. An alternative embodiment of a marking jig **800** may have a different number of radial arms and marking pads than are shown in the figures.

[0091] FIG. **29** shows an example of trocar entry point markings made on the exterior surface of an eye by the marking instrument **800** of FIGS. **26-28**. In the example of FIG. **29**, the iris **1006** is represented as a shaded area between the edge of the pupil **1016** and the limbus **1008**, with Schlemm's canal **1010** close to the limbus. The limbus **1008**, Schlemm's canal **1010**, and edge of the pupil **1016** are represented in FIG. **29** by dashed lines to distinguish these lines from markings made by the instrument **800**. The sclera **1002** is represented in FIG. **29** by areas outside the perimeter of the limbus **1008**.

[0092] Each pair of marking pads **810**, **812** prints a corresponding pair of line segments **818**, **820** tangent to Schlemm's canal **1010**. Each pair of line segments **818**, **820** meets at an intersection point **822** corresponding to a trocar entry point **606** on the sclera. More than one entry point **606** may be marked to provide a choice of trocar insertion points for reaching a target area in an eye. Each intersection point **822** is offset in a radial direction from the limbus **1008** by the predetermined separation distance **826**. A dot or other marking may be placed at each intersection point **822** to enhance visibility of the positions of the trocar entry points **606**.

[0093] After the trocar entry point marking instrument **800** transfers the pattern of intersection line segments from the example of FIG. **29** to the surface of the eye, one of the intersection points **822** may be selected for insertion of the trocar through the sclera. The distal end **210** of the trocar **200** is preferably placed in direct contact with the intersection point **822** on the sclera. The shaft **206** of the trocar is preferably made parallel to one of the line segments **818**, **820** and the trocar advanced parallel to the line segment in the direction from the intersection point **822** to the limbus **1008** until the illuminated distal end **220** of the trocar is observed to enter Schlemm's canal **1010**. After the trocar enters Schlemm's canal, the composite microcannula **300** may be extended from the end of the trocar as described for the examples of FIGS. **21** and **22**.

[0094] FIG. **30** shows an example of some steps included with a method embodiment. A method in accord with an embodiment **700** may include any one or more of the following steps, in any combination: [0095] at step **702**, placing a distal end of a composite microcannula within a lumen of a trocar; and [0096] at step **704**, selecting a structure in an eye for receiving the trocar. Schlemm's canal, a collector channel, and a blood vessel are examples of structures which may be selected, but it will be appreciated that embodiments of the microsurgical instrument **100** may be used to introduce a composite microcannula into other chambers, vessels, or channels in an eye or another organ.

[0097] The example of a method embodiment may further include: [0098] at step **706**, selecting a trocar entry point on an eye; [0099] at optional step **708**, marking a trocar entry point with a marking jig, for example the trocar entry point marking instrument **800** in the example of FIG. **26**; [0100] at step **710**, illuminating the distal end of the composite microcannula, thereby illuminating the distal end of the trocar; [0101] at step **712**, positioning the trocar at the selected trocar entry point; [0102] at step **714**, advancing the trocar from the selected trocar entry point until the illuminated distal end of the trocar is observed to enter the selected structure in the eye; and [0103] at step **716**, extending the composite microcannula from the distal end of the trocar toward a target

region in the eye, thereby transitioning from illuminating the distal end of the trocar to illuminating tissue outside the trocar. Examples of a target region include, but are not limited to, a chamber, vessel, channel or canal blocked by obstructing material, and a collapsed or constricted space to be enlarged or re-opened.

[0104] The example method embodiment may optionally include any one or more of: [0105] holding the distal end of the composite microcannula stationary relative to the trocar; [0106] selecting Schlemm's canal as the structure to be entered by the illuminated distal end of the trocar; [0107] advancing the trocar from the selected trocar entry point along a line tangent to Schlemm's canal; [0108] marking the trocar entry point at the intersection of two lines, each of the two lines tangent to Schlemm's canal; [0109] positioning the two tangent lines to intersect a preferred distance from the limbus of an eye; [0110] centering a marking instrument over the pupil of an eye and pressing the marking instrument against the eye to mark the two tangent lines onto the surface of the eye; [0111] marking more than one trocar entry point each time the marking instrument is pressed against the eye; [0112] illuminating the distal end of the trocar with another light guide; [0113] inserting a payload into the composite microcannula and delivering the payload to the target region; [0114] illuminating the distal end of the trocar with electromagnetic radiation having a wavelength not visible to unaided human vision and observing the distal end of the trocar with a camera sensitive to the electromagnetic radiation; [0115] withdrawing the composite microcannula through the trocar while keeping the trocar at the trocar entry point; [0116] inserting a payload into the composite microcannula; [0117] moving the payload through the composite microcannula to the target region; [0118] retracting the composite microcannula with the payload remaining in the target region; and [0119] keeping the trocar stationary relative to the eye after the distal end of the trocar enters the selected structure in the eye.

[0120] FIGS. **34**, **35**, and **36** show another example of a microsurgical instrument having a positioner configured for extending and retracting a composite microcannula through the lumen of a trocar. The positioner **400** is configured for attachment of a trocar **200** to the distal end **418** of a handpiece **420**. The composite microcannula **300** passes through the lumen of the trocar **300** and through the handpiece **420**, where the microcannula is coupled to an insert **488** slidably engaged with the handpiece **420**. As will be explained in more detail below, sliding an actuator **424** in a slot **428** on the handpiece **420** by a selected distance **484** causes a corresponding displacement of the composite microcannula with respect to the end of the trocar's rigid hollow shaft **206** by a distance **486** equal to twice the displacement distance **484** of the actuator. The trocar **200** and the composite microcannula **300** may be provided separately from the positioner **400** and may be removable for replacement should either the trocar or microcannula become damaged, contaminated, or otherwise unusable for a particular procedure.

[0121] Some internal features of the example positioner **400** of FIG. **34** are shown in two cross-sectional views in FIG. **36**. Section H-H in FIG. **36** shows the actuator **424** retracted toward the proximal end **416** of the handpiece **420**. Section J-J shows the actuator advanced toward the distal end **418** of the handpiece **420**. The two views of the handpiece **420** are referenced to one another in FIG. **36** so that the positions of internal components may be accurately compared between the cross-sections.

[0122] The insert **488** is slidably engaged with the interior surface of a void formed inside the handpiece **420**. The actuator **424** travels along one or more guide ridges **430**. A U-shaped hollow tube **492** attached to the actuator **424** moves with the actuator. An insert displacement wire **490** passes slidably through the lumen of the U-shaped hollow tube **492**. One end of the insert displacement wire **490** is affixed to the insert **488**. The opposite end of the insert displacement wire **490** is affixed to an anchor post **494** strongly attached to, or alternately formed as an integral part of, the handpiece **420**. The insert displacement wire **490** is preferably flexible enough to slide easily around the bend in the U-shaped hollow tube **492**, yet stiff enough to push the insert **488** in a distal direction (i.e., away from the trocar **200**) when the actuator is displaced in a distal direction.

[0123] With one end of the insert displacement wire **490** fixed to the handpiece at the anchor post **494** and the other end of the wire fixed to the insert **488**, a sliding displacement of the actuator **424** along the handpiece by a distance “d” **484** causes the U-shaped hollow tube **492** to be displaced by the same distance “d” and the insert **488** to be displaced by twice as much ($2 \times d$), represented in the figures by a relative displacement **486** of the insert between cross sections H-H and K-K. The composite microcannula **300** is coupled to the insert **488** sufficiently strongly to keep the composite microcannula stationary relative to the insert when the actuator is moved relative to the handpiece. Because the insert moves twice the displacement distance **484** of the actuator, the composite microcannula also moves twice the displacement distance **484** of the actuator. Moving the actuator proximally by a distance “d” **484** extends the composite microcannula from the trocar by a distance “ $2 \times d$ ” **486**. Moving the actuator distally by a distance “d” retracts the composite microcannula by a distance “ $2 \times d$ ”.

[0124] The composite microcannula **300** may pass through a hollow sleeve having a fixed segment **436** attached to, or alternately formed as an integral part of, the rigid hollow shaft **206** of the trocar **200**. A movable segment **434** of the hollow sleeve is slidably engaged with the fixed segment **436** at one end and attached to the insert **488** at the other end. The hollow sleeve limits transverse deflection of the composite microcannula as the actuator is advanced and retracted, forcing the composite microcannula to extend and retract without kinking or significant bending inside the handpiece.

[0125] As suggested in the example of FIG. **36**, the composite microcannula **300** may extend outward from the proximal end **416** of the positioner **400** and insert **488**. The proximally-extending portion of the composite microcannula may be configured to receive solid and/or liquid payloads, surgical instruments, and one or more optical fibers as previously described.

[0126] Unless expressly stated otherwise herein, ordinary terms have their corresponding ordinary meanings within the respective contexts of their presentations, and ordinary terms of art have their corresponding regular meanings.

Claims

1. An apparatus for use in ophthalmic surgery, the apparatus comprising: a handpiece; a trocar attached to the handpiece, the trocar having a lumen; wherein the trocar is configured to reduce abrasion or cutting of objects passing through the lumen; the lumen having a microcannula passing therethrough; the microcannula comprising a fluid communication channel for delivery of a payload; and wherein the payload is able to be loaded into the handpiece prior to use.
2. The apparatus of claim 1, wherein the trocar comprises a smoothed distal edge configured to reduce abrasion or cutting of objects which pass through the lumen.
3. The apparatus of claim 1, further comprising an optically transmissive element configured to transmit light from a light source, wherein the light source is disposed within the apparatus.
4. The apparatus of claim 3, wherein the apparatus is configured to be used as a single handed, single operator-controlled apparatus.
5. The apparatus of claim 1, where the microcannula can be advanced about a 180-degree arc of Schlemm's Canal.
6. The apparatus of claim 1, where the microcannula can be advanced about a 360-degree arc of Schlemm's Canal.
7. The apparatus of claim 5, wherein the microcannula is configured to deliver a viscoelastic material.
8. The apparatus of claim 5, where the microcannula is configured to deliver a payload from the group consisting of particles, nano particles, pharmaceuticals, small devices, a microsurgery instrument, and a stent.
9. An apparatus for use in ophthalmic surgery, the apparatus comprising: a handpiece; an actuator

coupled to the handpiece; an insert slidably coupled to the handpiece; a trocar attached to the handpiece; wherein the trocar is configured to reduce abrasion or cutting of objects passing through a lumen of the trocar; and a microcannula passing through the lumen and coupled to the insert; wherein a displacement distance of the insert relative to the actuator corresponding to a length of the portion of the microcannula extending outward from a distal end of the trocar to a distance of a movement of the actuator.

10. The apparatus of claim 9, wherein the microcannula comprises one or more proximal ends and a distal end, further comprising one of more communicating elements extending from the one or more proximal end to the distal end.

11. The apparatus of claim 10, comprising a plurality of communicating elements, wherein a proximal end of the plurality of communicating elements are of different lengths.

12. The apparatus of claim 11, wherein the plurality of communicating elements are configured for fluid delivery and/or payload delivery and/or illumination.

13. The apparatus of claim 10 wherein the plurality of communicating elements are configured to be detachably connected to one or more external source before use.

14. The apparatus of claim 9 where the corresponding distance is not limited to being 1:1.

15. An apparatus, comprising: a handpiece; a trocar attached to the handpiece; wherein the trocar is configured to reduce abrasion or cutting of objects passing through a lumen of the trocar; and a microcannula passing through the trocar; wherein the microcannula comprises a fluid or payload delivery element; and wherein the microcannula further comprises an optically transmissive element configured to transmit light from a light source, wherein the light source is disposed within the apparatus.

16. The apparatus of claim 15, wherein the handpiece or microcannula is configured to load a fluid or payload prior to use.

17. The apparatus of claim 15, wherein the handpiece or microcannula is configured to load a fluid or payload while in use.

18. The apparatus of claim 15, wherein the handpiece comprises an actuator configured to control the microcannula; and wherein the displacement distance of the microcannula corresponding to a length of the portion of the microcannula extending outward from a distal end of the trocar to a distance of a movement of the actuator.

19. The apparatus of claim 16, wherein the apparatus is configured to be used as a single handed, single operator-controlled apparatus.

20. The apparatus of claim 16, wherein the microcannula can be advanced about a 360-degree arc of Schlemm's Canal.
